

DATA ANALYTICS

CATALYST
SKILL HUB
TECH-HUB

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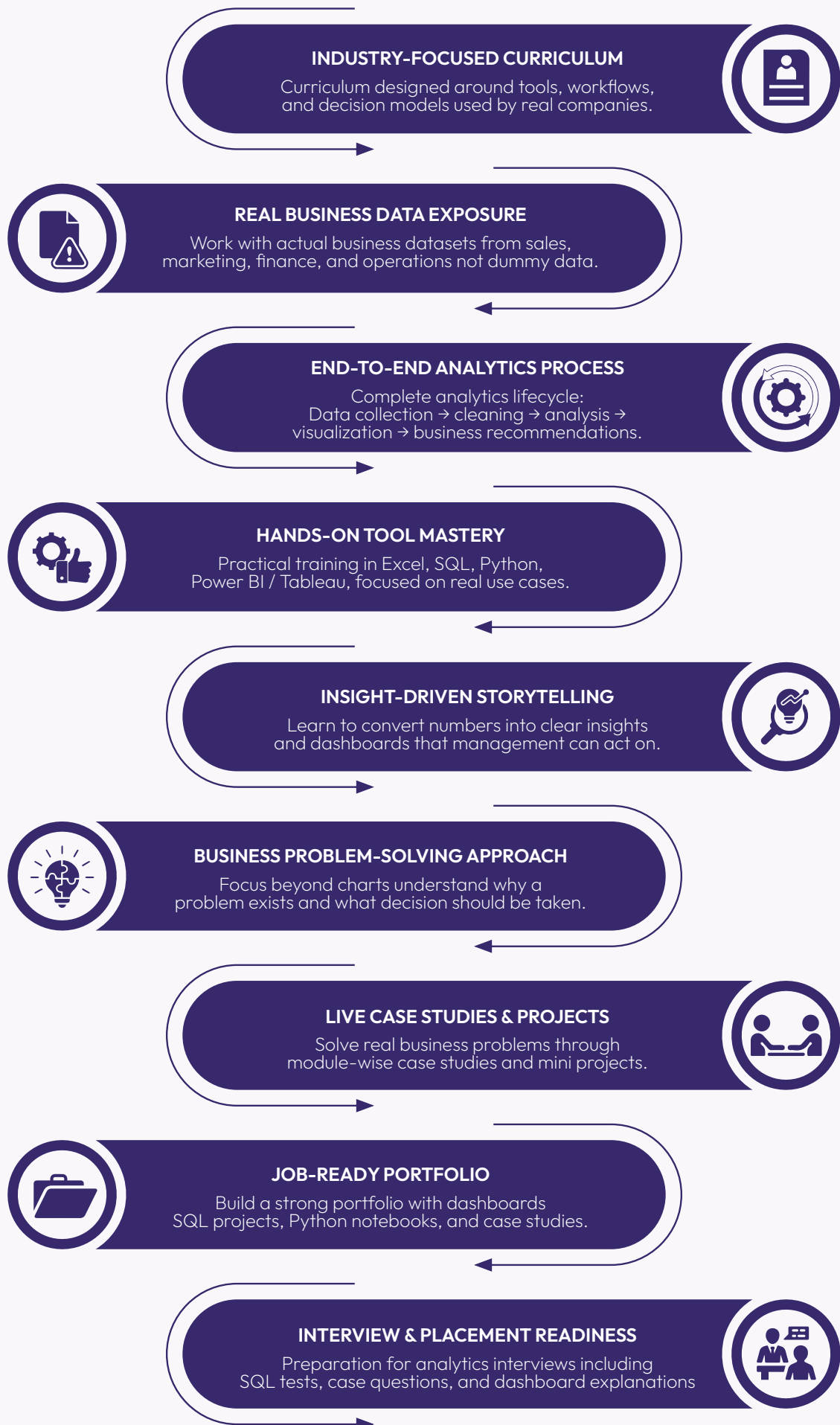
MISSION

At Catalyst, our mission is to redefine coding education by focusing on active, practical, and mentor-driven learning. We aim to equip aspiring software developers with strong technical skills, real-world problem-solving abilities, and the confidence needed to thrive in the fast-changing tech industry. Through a hands-on approach, expert guidance, and a supportive learning environment, we are committed to expanding career opportunities and helping students achieve self-sufficiency in their professional journeys.

VISION

Catalyst aspires to become a leading tech learning hub in Kerala and beyond, known for empowering confident, skilled developers who can adapt to the changing demands of the tech industry. We aim to expand job opportunities, help students achieve self-sufficiency, and equip them to take on leadership roles in the tech industry, ultimately shaping the future of technology and making a lasting impact on the global tech ecosystem through innovative education and mentorship.

DATA ANALYTICS – CORE FEATURES



Data Analytics Course Curriculum

Module 1

Introduction To Data Analytics

1. What is Data Analytics?

- Definition of data, information, and insights
- Difference between raw data and processed data
- Importance of data in modern businesses
- Examples from real industries

2. Types of Analytics

- Descriptive Analytics (reports, summaries)
- Diagnostic Analytics (root cause analysis)
- Predictive Analytics (forecasting models)
- Prescriptive Analytics (decision systems)
- Tools used for each type

3. Data Analytics Lifecycle

- Problem identification
- Data collection methods
- Data preparation
- Data analysis
- Visualization
- Decision making

4. Roles & Responsibilities

- Data Analyst duties
- Business Analyst duties
- BI Developer duties
- Data Scientist overview
- Communication skills

5. Industry Use Cases

- Healthcare analytics
- Banking & finance analytics
- Marketing analytics
- HR analytics
- E-commerce analytics

Module 2

Business & Analytical Thinking

1. Understanding Business Problems

- Identifying business objectives
- Understanding stakeholders
- Business requirement documents
- Problem statements
- Scope definition

2. KPIs & Metrics

- Financial KPIs
- Sales KPIs
- Marketing KPIs
- Operational KPIs
- HR KPIs
- Performance dashboards

3. Data-Driven Decision Making

- Evidence-based decisions
- Trend analysis
- Benchmarking
- Risk assessment
- Scenario analysis

4. Case Studies

- Sales improvement cases
- Customer retention cases
- Cost reduction cases
- Market expansion cases
- Product performance analysis

Module 3

Statistics for Data Analytics

1. Types of Data

- Qualitative vs Quantitative
- Discrete vs Continuous
- Structured vs Unstructured
- Primary vs Secondary data

2. Descriptive Statistics
Mean, Median, Mode
Range and Quartiles
Variance
Standard deviation
Skewness
Kurtosis

3. Probability Basics
Probability rules
Events and outcomes
Conditional probability
Bayes theorem (intro)
Random variables

4. Correlation & Regression
Pearson correlation
Scatter plots
Simple linear regression
Regression line
Interpretation of results

5. Hypothesis Testing (Intro)
Null and alternative hypothesis
Z-test and T-test
P-value
Significance level
Decision rules

Module 4

Advanced Excel for Data Analytics

1. Data Cleaning & Preprocessing
Remove duplicates
Text to columns
Find and replace
Data validation
Handling missing values

2. Excel Formulas & Functions

- Logical functions (IF, AND, OR)
- Lookup functions (VLOOKUP, XLOOKUP)
- Math functions (SUM, ROUND)
- Statistical functions
- Text functions

3. Pivot Tables & Charts

- Creating pivot tables
- Grouping data
- Calculated fields
- Pivot charts
- Slicers

4. Conditional Formatting

- Highlight rules
- Color scales
- Icon sets
- Custom formulas

5. Dashboard Creation

- Layout design
- Charts selection
- Interactivity
- Linking pivots
- Report automation

Module 5

SQL for Data Analytics (Basics)

1. Database Fundamentals

- Database types
- Tables and schemas
- Primary keys
- Foreign keys
- Relationships

2. SELECT Statements

- Column selection
- DISTINCT keyword
- Aliases
- Expressions
- Calculated columns

3. WHERE, ORDER BY

- Filtering operators
- Logical operators
- Pattern matching (LIKE)
- Sorting techniques
- Limiting records

4. Aggregate Functions

- COUNT, SUM, AVG, MIN, MAX

5. GROUP BY, HAVING

- Grouping logic
- Filtering groups
- Multiple columns
- Conditional groups
- Performance tips

6. Simple Joins (Intro)

- INNER JOIN, LEFT JOIN, RIGHT JOIN
- Table relationships
- Join conditions

7. Real-Time Queries

- Business reports
- Live dashboards
- Case queries
- Practice datasets

Module 6

Python Basics

1. Python Syntax

- Variables, Keywords, Indentation
- Comments
- Naming rules

2. Data Types
int, float, string, list, tuple, dictionary, set
3. Control Statements
if-else, nested conditions
for loop, while loop, break and continue

4. Functions
Defining functions
Parameters
Return values
Lambda functions
Recursion (intro)

5. File Handling Basics
Reading files, Writing files
CSV handling
Error handling
File paths

Module 7

Python Libraries for Data Analytics

1. NumPy
Array creation
Indexing and slicing
Mathematical operations
Statistical functions
2. Pandas
Series creation, DataFrame creation
Importing files, Exporting data, Indexing methods
3. Matplotlib & Seaborn
Line plots, Bar charts, Histograms, Scatter plots, Subplots
Customizing labels and titles, Statistical plots, Distribution plots (distplot, kde)
Categorical plots (box, violin, count), Pair plots, Heatmaps,
Theme and style settings

4. Data Cleaning & Transformation

- Handling missing values
- Removing duplicates
- Data type conversion
- Renaming columns
- Feature engineering

5. Exploratory Data Analysis (EDA)

- Summary statistics
- Distribution analysis
- Outlier detection
- Correlation analysis
- Pattern discovery

6. Data Visualization

- Line charts, Bar charts, Histograms, Scatter plots, Heatmaps

Module 8

Data Visualization Concepts

1. Principles of Visualization

- Simplicity, Accuracy, Consistency, Clarity, Relevance

2. Choosing the Right Charts

- Comparison charts, Trend charts, Distribution charts, Relationship chart, Composition charts

3. Storytelling with Data

4. Business Reporting Standards

- Report templates, KPI dashboards, Presentation formats

Module 9

Git & GitHub

- Version control concepts

- Git commands

- Branching & merging

- GitHub repositories

- Collaboration workflow

Module 10

Power BI – Basics

1. Power BI Ecosystem
Desktop, Service, Mobile, Gateways
2. Data Import
Excel fi les, CSV fi les, Databases, Web sources, APIs
3. Power Query
Data cleaning, Transformations, Merge queries, Append queries, Parameters
4. Data Modeling
Relationships, Star schema, Cardinality
5. Basic Visuals
Tables, Cards, Bar charts, Line charts, Slicers

Module 11

Power BI – Advanced

1. DAX Fundamentals
Syntax, Operators, Context, Variables, Formatting
2. Measures & Calculated Columns
Difference, Performance tuning
Dynamic measures, KPI, Debugging
3. Advanced Visuals
Custom visuals, Maps, Decomposition tree, AI visuals, Matrix visuals

- 4. Interactive Dashboards
Drill down, Drill through, Bookmarks, Tooltips

- 5. Publishing Reports
Power BI Service, Workspace, Sharing, Security

Module 12

Real-World Projects

- 1. Excel Project
 - Sales analysis
 - Financial modeling
 - Dashboard creation
 - Automation
 - Reporting
- 2. SQL Mini Project
 - Database design
 - Query writing
 - Performance tuning
 - Report generation
 - Documentation
- 3. Python Analysis Project
 - Data collection, Cleaning, EDA, Visualization, Reporting
- 4. Power BI Dashboard Project
 - Data modeling
 - DAX measures
 - Interactive reports
 - Publishing
 - Feedback
- 5. Business Case Study
 - Problem analysis
 - Data mapping
 - Solution design
 - Presentation
 - Evaluation

GIT & VERSION CONTROL PRESENCE

Active GitHub Profile:

Maintain a consistent contribution graph with regular commits

Open Source Exposure (Optional):

- Contribute to beginner-friendly repositories
- Fix small issues or documentation

Git & Version Control Presence

Repository Quality:

Well-structured repositories with

- Clean folder structure
- Meaningful commit messages
- Proper README documentation

Active GitHub Profile:

Maintain a consistent contribution graph with regular commits

Project Documentation:

Each project should include

- Project overview
- Features list
- Tech stack used
- Installation steps
- Screenshots or demo GIFs

Repository Quality:

Well-structured repositories with:

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Commit Discipline:

Follow best practices like:

- Feature-based commits
- Branching strategy (main/dev/feature)
- Descriptive commit messages (not "update" or "final")

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Portfolio Development Strategy

Personal Portfolio Website:

Create a responsive portfolio showcasing:
About Me section (career journey + skills)
Projects section (detailed project cards)
Skills & Tech Stack
Contact section

Project Presentation Format:

For each project include:

- Problem statement
- Your solution
- Key features
- Technologies used
- Challenges faced
- Live demo link + GitHub link

Visual Proof:

- Screenshots of dashboards
- Admin panels
- Mobile responsiveness

Deployment Practice:

- Host projects on platforms like Vercel / Netlify / Render
- Connect custom domain (optional but impressive)

Performance & UX:

- Fast loading
- Mobile-friendly design
- Easy navigation

Resume + Portfolio Integration:

- Add portfolio link in LinkedIn
- Add GitHub link in resume
- Add resume download option in portfolio

LINKDIN MARKETING

LinkedIn Profile Transformation

- Professional Branding: We'll optimize your profile with industry-specific keywords (Java/Python/Full-stack/Cloud)
- Headline Engineering: Craft compelling headlines like "Full Stack Developer | Java | React | AWS | Seeking Opportunities"
- Summary Makeover: Create achievement-driven summaries showcasing projects and technical skills

Content & Engagement Strategy

- Project Showcases: Regular posts featuring your GitHub projects with:
 - Architecture diagrams
 - Problem-solution narratives
 - Live demo links
- Learning Journey: Document your skill progression with:
 - "What I learned this week" posts
 - Code snippet explanations
 - Tech stack comparisons

Network Building

- Strategic Connections: Connect with:
 - Tech recruiters (500+ target)
 - Hiring managers in target companies
 - Alumni from our institute
 - Industry influencers
- Company Following: Follow 50+ target companies for job alerts and industry insights

Job Hunting Optimization

- Open-to-Work Activation: Use LinkedIn's green banner strategically
- Keyword Optimization: Profile tuned for ATS and recruiter searches
- Recommendations: Secure 3+ recommendations from instructors and peers

LinkedIn Learning Integration

- Badge Display: Showcase completed LinkedIn Learning courses
- Skill Assessments: Take and pass 5+ technical skill assessments
- Certifications: Highlight completed certifications on your profile

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